

Dynamic Annuity-Rate Model -- Fit Report

Empirical fit of the interest-rate process used by the dynamic Monte Carlo mode. Regenerate with `python fit_rate_model.py --markdown`.

1. Purpose

In the dynamic simulation mode the annuity discount rate is not fixed; it evolves with inflation through the empirical link between the **10-year Treasury yield** and **inflation**. Both are persistent, correlated time series, so the rate is a partial adjustment toward a long-run Fisher target driven by *lagged* inflation:

$$i^*_t = a + b * \pi_t(t-1)$$

(long-run Fisher target)

$$i_t = (1 - \text{lam}) * i_t(t-1) + \text{lam} * i^*_t + \text{eps}_t$$

(error correction)

`a` is the real-rate intercept, `b` the long-run inflation pass-through, and `lam` the annual adjustment speed; `eps_t ~ N(0, sigma^2)`.

2. Data and sample

Dependent variable	10-year Treasury constant-maturity yield, annual average of monthly FRED GS10 [1]
Inflation regressor	CPI-U annual-average inflation (BLS, via usinflationcalculator.com) [2]
Sample	1954-2025 (n = 72 annual observations)
Estimation	Ordinary least squares (reduced form), <code>numpy.linalg.lstsq</code>

GS10 begins in 1953; the lagged regressors drop the first year, so estimation starts in 1954. The yield series lives in `market_data.py` (`treasury_10y_regression_data()`); the regression is `rate_model.fit_interest_model()`.

3. Reduced-form regression

$$i_t = c + \rho * i_t(t-1) + \delta * \pi_t(t-1) + \text{eps}_t$$

term	coef	std err	t
c (intercept)	0.00279	0.00202	1.38

term	coef	std err	t
$\rho \cdot i_{(t-1)}$	0.83457	0.04242	19.67
$\delta \cdot \pi_{(t-1)}$	0.18661	0.04482	4.16

$R^2 = \mathbf{0.9267}$, adjusted $R^2 = 0.9246$, residual sigma = **0.00786 (0.79%)**, Durbin-Watson = **1.945**.

The high `rho` (0.835, $t = 19.7$) confirms the strong persistence of yields; the significant `delta` (0.187, $t = 4.2$) is the inflation pass-through. The Durbin-Watson near 2 means the residuals show little remaining first-order autocorrelation, so the AR(1)-in- levels specification is adequate. The intercept `c` is not individually significant ($t = 1.38$), which is expected -- the long-run real rate is weakly identified separately from the persistence term.

4. Structural parameters

Mapping `lam = 1 - rho`, `a = c / lam`, `b = delta / lam`, with `sigma` the residual standard deviation:

parameter	symbol	value
adjustment speed	lam	0.1654
real-rate intercept	a	0.01688 (1.69%)
inflation pass-through	b	1.1280
annual rate shock	sigma	0.00786 (0.79%)

These are the values `rate_model.py` bakes in as `DEFAULT_ADJUST_SPEED`, `DEFAULT_REAL_INTERCEPT`, `DEFAULT_INFLATION_BETA`, and `DEFAULT_RATE_SIGMA` (computed at import from the same data, so the simulator and this report cannot drift apart). The simulated rate path starts from a current-level initial rate, `DEFAULT_INITIAL_RATE = 4.3%` (the 2025 GS10 annual average), and is floored at 0%.

5. Interpretation

- **Long-run pass-through** `b = 1.13` -- slightly above one-for-one, consistent with the long-run Fisher relationship (and the tax-augmented Darby variant, which predicts a coefficient above unity) [3].
- **Adjustment speed** `lam = 0.17` -- only about 17% of the gap to the Fisher target closes each year, so the rate is **sluggish**. When inflation jumps, the nominal rate lags and the *implied real rate* (`i - pi`) compresses and can turn negative -- the Mundell-Tobin / short-run "Fisher puzzle" behaviour actually seen in 2021-2022 [4]. The long-run real rate settles near `a + (b-1)·pi`: about **2.01%** at 2.5% inflation.

- **Steady state.** At a constant 2.5% inflation the rate converges to $a + b \cdot 0.025 = \mathbf{4.51\%}$, close to the ~5% level at which the local annuity pricer matches market quotes, and to the recent 10-year yield.
- **Shock size** `sigma` = **0.79%** -- the typical one-year surprise in the rate beyond what persistence and inflation explain.

6. Scope and caveats

- Per the *layered* design, only the **interest** equation is fit here. Inflation itself is not parametrically modelled: in simulation it comes from the block bootstrap of historical equity/inflation pairs (`equity_model.py`), so there is no separate inflation AR to estimate.
- The rate innovation `eps_t` is drawn independently of the equity/inflation innovations (a simplification; their historical residual correlation is small).
- The fit uses the GS10 sample (1954-2025); a different yield maturity, frequency (year-end vs. average), or window would shift the estimates. The series and retrieval are pinned in `market_data.py`.

References

1. Federal Reserve Bank of St. Louis, FRED, *Market Yield on U.S. Treasury Securities at 10-Year Constant Maturity (GS10)*. <https://fred.stlouisfed.org/series/GS10>
2. U.S. Bureau of Labor Statistics, *CPI-U*, via *Historical Inflation Rates*. <https://www.usinflationcalculator.com/inflation/historical-inflation-rates/>
3. Fisher, I. (1930). *The Theory of Interest*. Macmillan. (Darby, M. R. (1975), *The Financial and Tax Effects of Monetary Policy on Interest Rates*, for the above-unity tax-augmented coefficient.)
4. Mundell, R. (1963). *Inflation and Real Interest*. *Journal of Political Economy*, 71(3); Tobin, J. (1965). *Money and Economic Growth*. *Econometrica*, 33(4).